Finding a Topic, Formulating a Research Question, and Thinking About a Design

Digital Methods Lecture 2

Course Responsible: Hjalmar Bang Carlsen, Associate Professor, SODAS. hc@sodas.ku.dk

Today's Tasks

1. Generate topics
2. Formulate research questions
3. Prototype a design to answer the research question
    - The role of qualitative sub-studies

What is a Topic?

- Simply a device to concentrate your attention on some things and not others.
- You can use heuristics to rearrange elements in order to view your topic from a new perspective.

Topic Ideas, Please!

What is a Topic?

- A broad theme
- An event
- A group
- An activity
- An idea/discourse
- A technology
- A location

Framing My Topic as X

- A broad theme
- An event
- A group
- An activity
- An idea/discourse
- A technology
- A location

Theoretical and Empirical Casing

- Some begin with a very empirical topic and need to conceptualize it.
- Others start with a theoretical topic and need to approach it empirically.

Casing: What is an (Empirical) Topic a Case of?

- Casing — from an empirical to a theoretical case
- Casing determines literature
- Casing influences the relevant questions

What is X a Case of?

In groups, take either your case or another case proposed and translate it into 2 or 3 different theoretical cases.

Case Selection: What Makes an Empirical Case Good for My Topic?

- Casing from a theoretical to an empirical case (case selection).
- In the universe of possible cases, what is your case good or bad for?
- Case selection has analytical consequences.

Coming Up with a Question

Some Questionable Rules for Questions

General Rules:

- You should be able to answer your question.
- You should learn something after answering your question.
- You should not know the answer beforehand.
- Your first question formulation is not your final formulation.

Some Questionable Rules for Questions

Mixed Rules:

- Some aspects of your question(s) need to be answerable with both quantitative and qualitative methods.
- The same question can be answered using both qualitative and quantitative evidence.

Types of Questions 1

- Theory-testing questions
- Theory-generating questions
- Descriptive questions
- Causal questions

	Theory Testing	**Theory Generating**
Descriptive	Is X characterised by Y? Does X correlate with Y?	What is X characterised by? What does X correlate with?
Causal	Does a rise in X cause a rise in Y?	What are the causes of Y? How does X cause Y?

Questions and Quant/Qual

- Qualitative analysis can be causal AND theory-testing.
- Quantitative analysis can be descriptive AND exploratory.

Types of Questions 2

- What (character)
- How (process)
- Where (setting)
- Why (causes/motivation)
- Who (actors)
- When (timing)

🔍 What – Descriptive / Definitional Questions

Purpose: To identify, define, or categorize a phenomenon.

Use when you want to know what something is, characterize a phenomenon, or identify the types of things that exist.

Examples:

- What are the components of incel culture?
- What frames do Ukrainian brigades use to mobilize support?
- What deservingness evaluations do Ukrainian refugee volunteers

🛠️ How – Process / Mechanism Questions

Purpose: To understand processes, mechanisms, or how things work together.

Use when you're interested in dynamics, sequencing, or relationships.

Examples:

- How do volunteers coordinate efforts in refugee solidarity groups?
- How do people with HIV navigate disclosure in daily life?

🌍 Where – Spatial / Contextual Questions

Purpose: To explore settings, locations, or spatial patterns.

Use when place, context, or institutional setting is relevant to your topic.

Examples:

- Where on Facebook do people discuss climate change?
- Where in Denmark do we find place-based stigmatization?
- Where in Denmark are there the most refugee solidarity groups?

❓ Why – Causal / Motivational Questions

Purpose: To explore reasons, causes, or motivations.

Use when you want to understand underlying forces or intentions.

Examples:

- Why do some volunteers stay active while others drop out?
- Why is there a gender gap in political participation online?
- Why do some users engage in hostile political commentary?

🧑 Who – Actor-Centric / Identity Questions

Purpose: To identify agents, actors, or social categories involved in a phenomenon.

Use when you're asking who is doing what, exploring social roles and identities, or examining the relative prevalence of actors.

Examples:

- Who participates in grassroots refugee support?
- Who participates in politics on Facebook?
- Who provides support in social networks?

⏰ When – Temporal / Timing Questions

Purpose: To explore the timing, duration, or historical emergence of phenomena.

Use when you're interested in sequences, timing, or change over time.

Examples:

- When are politicians responsive to social media feedback?
- When did politicians adopt Facebook?
- When did the concept of screen-time emerge in public discourse?

Exploiting Digital Data and Mixed Methods for Productive Questions

1. **New data for testing "old" theories**

> "[J]ust as the invention of the telescope revolutionized the study of the heavens, so too the technological revolution in mobile, Web, and Internet communications has the potential to revolutionize our understanding of ourselves and how we interact." (Watts, 2013)

Exploiting Digital Data and Mixed Methods for Productive Questions

1. New data for testing old theories
2. **New data for theory generation**

"Induced data structures can surprise, challenging presumptions or pre-existing theory, and lead the social analyst to abductively generate new theory by imagining what would be socially required for those patterns to exist." (Evans & Aceves, 2016)

Exploiting Digital Data and Mixed Methods for Productive Questions

1. New data for testing old theories
2. New data for theory generation
3. **Mixed digital methods for theory testing**

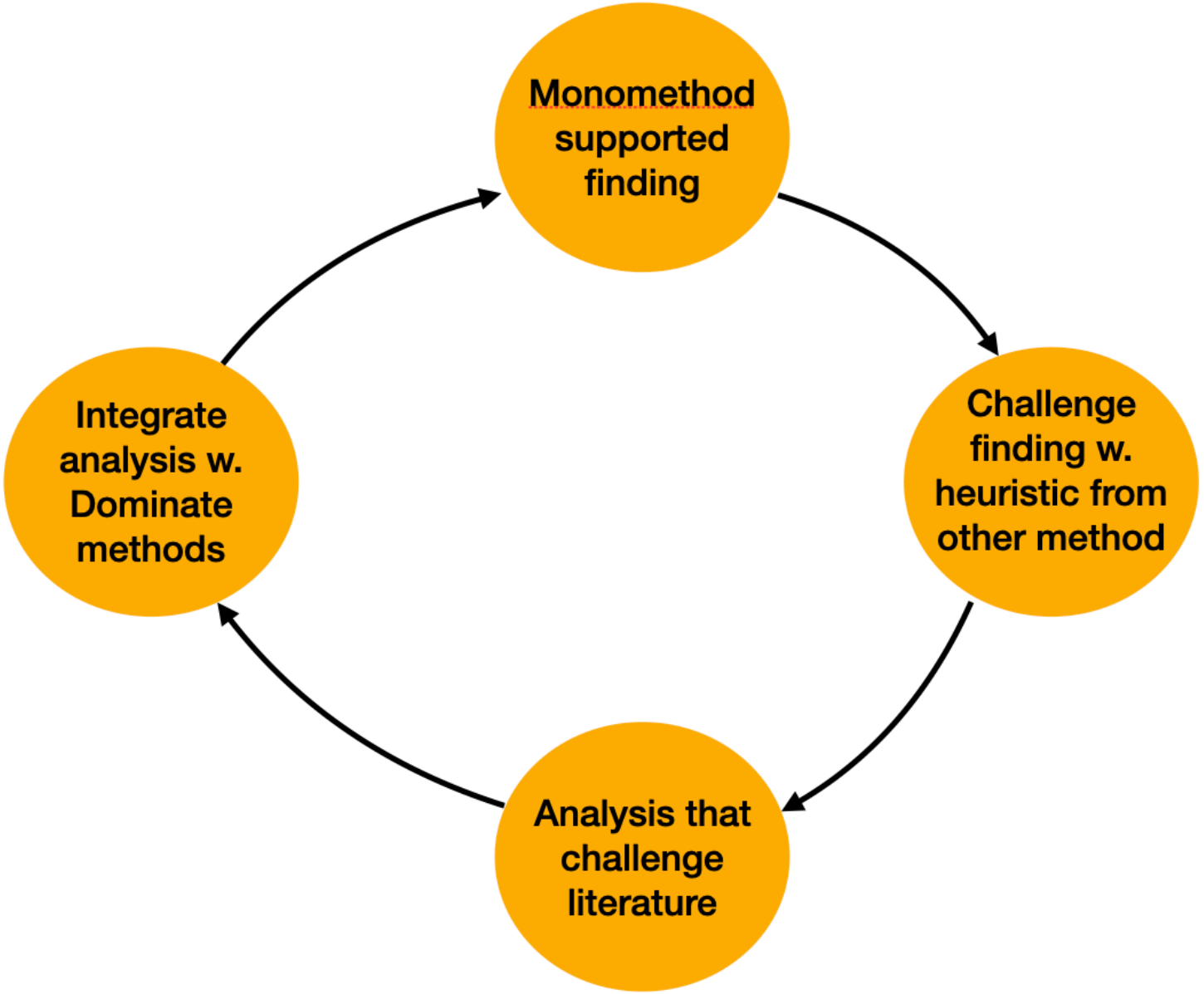
MM approaches help address the uncertainties of testing old theories with digital data.

Exploiting Digital Data and Mixed Methods for Productive Questions

1. New data for testing old theories
2. New data for theory generation
3. Mixed digital methods for theory testing
4. **Mixed digital methods for theory generation**

MM approaches support the discovery and grounding of your theories.

Mixed Method Motives


Monomethod supported finding
Challenge finding w. heuristic from other method
Analysis that challenge literature
Integrate analysis w. Dominate methods

Mixed Method Motives

- Motivated by the specific research question/problem
- Two overall justifications: confirmatory and complementary
  - **Confirmatory** studies seek to confirm or reject findings through the use of different types of data and analysis
  - **Complementary** studies use different data sources/analyses to answer different aspects of the research question.

Confirmatory Study Example

- Discrimination in hiring
- Combine surveys (explicit attitudes) and audit studies (actual behavior)

Complementary Study Example

- Relation between institutional trust and activism
- Survey (quantitative correlation) and interviews (mechanisms)

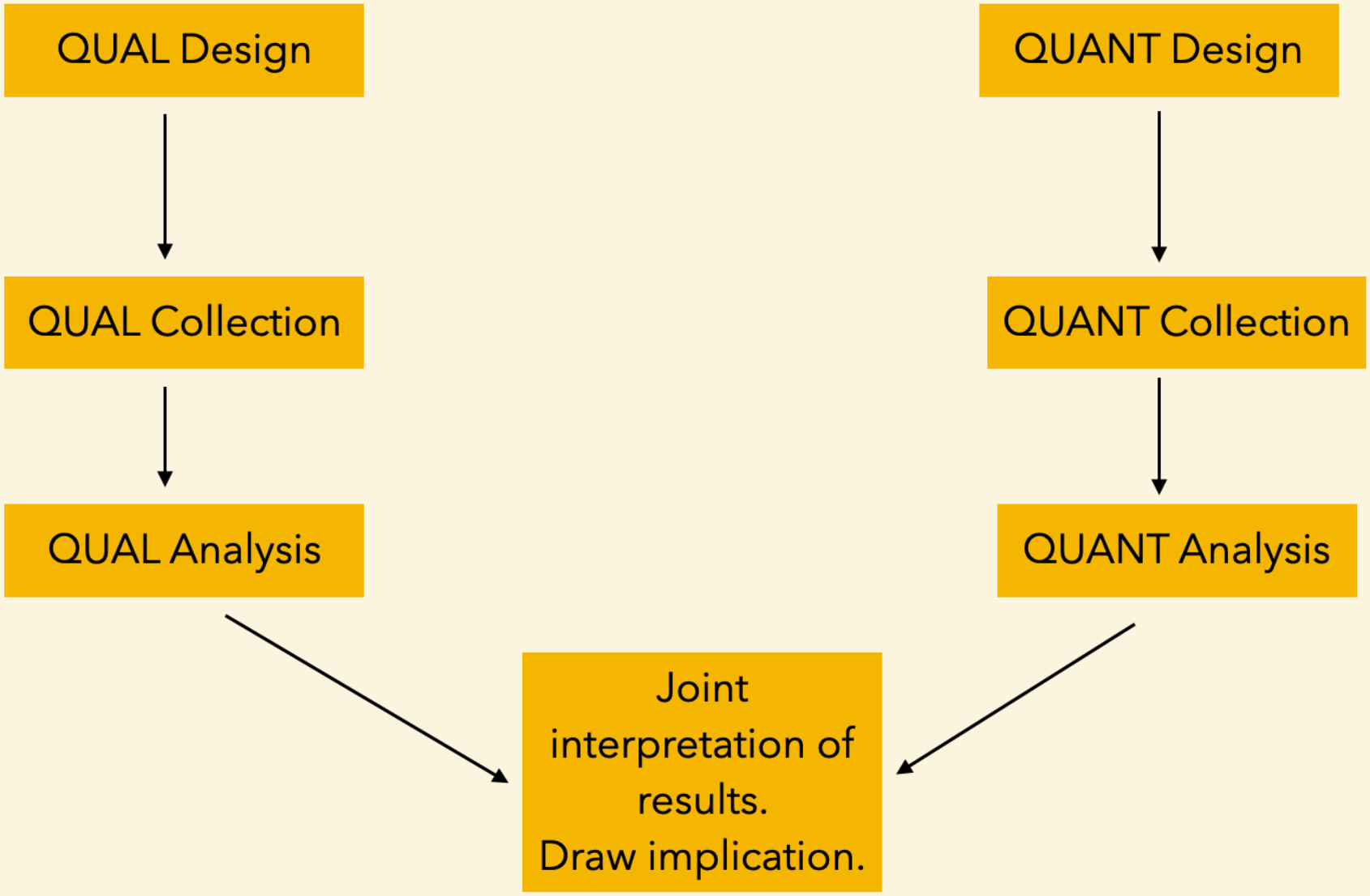
From Confirmatory and Complementary Designs to Operations

1. **Confirmatory Operations** - confirm patterns found with one method using another
2. **Complementary Operations** - use different methods to investigate different aspects.

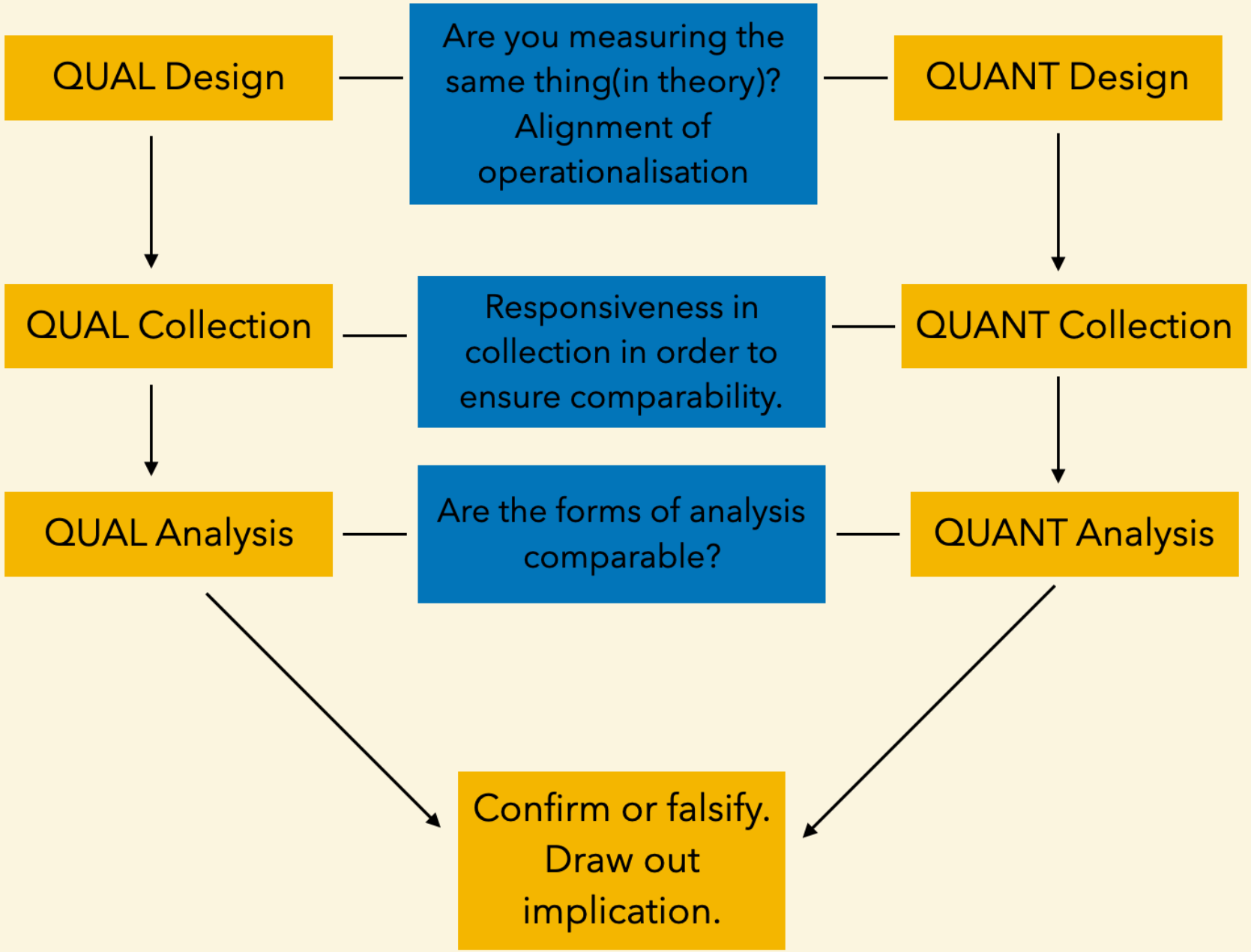
Sequencing of the design

- Concurrent design
  - Different studies are independent of one another
- Sequential design
  - Different studies are dependent upon one another

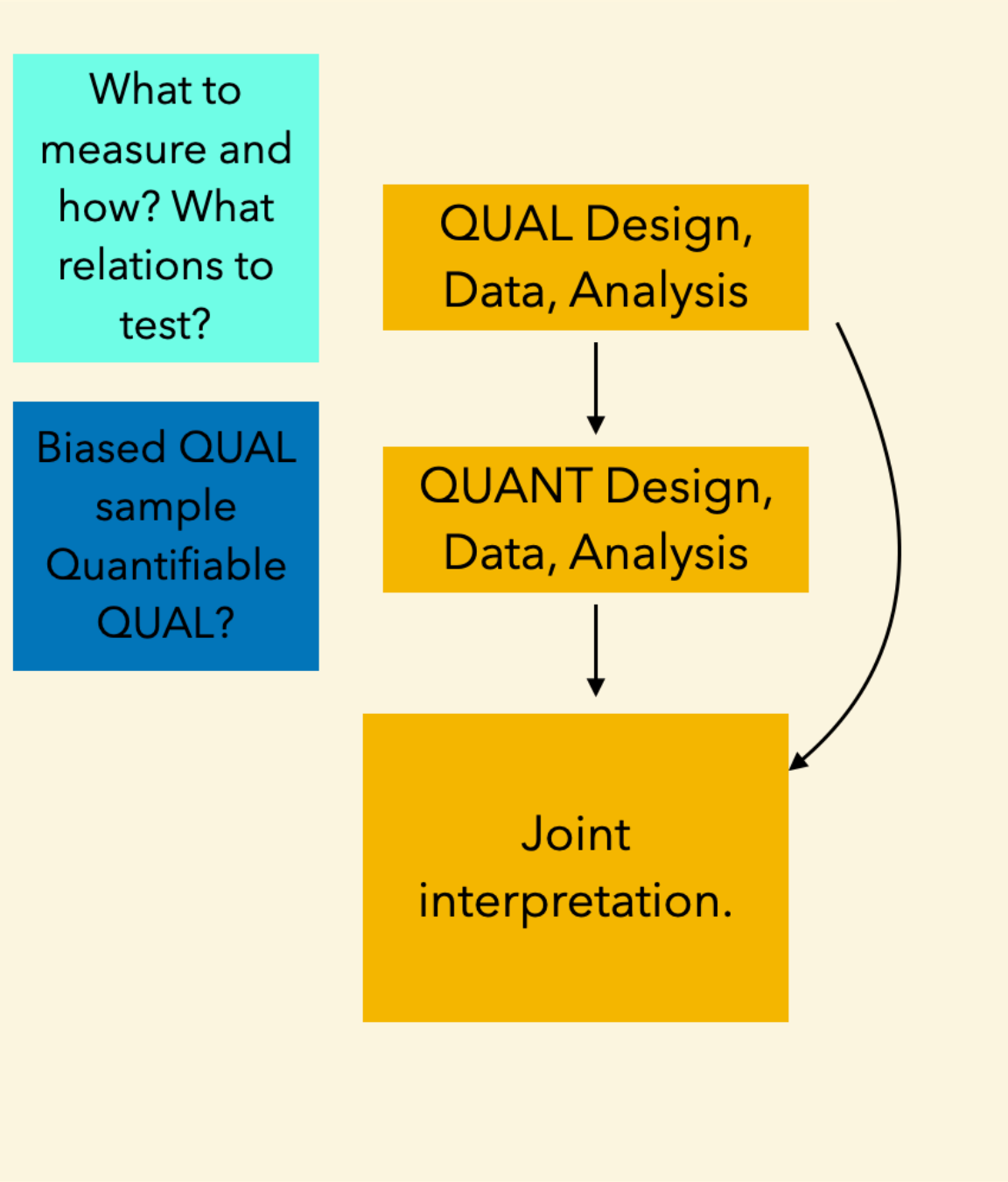
Con-current Design

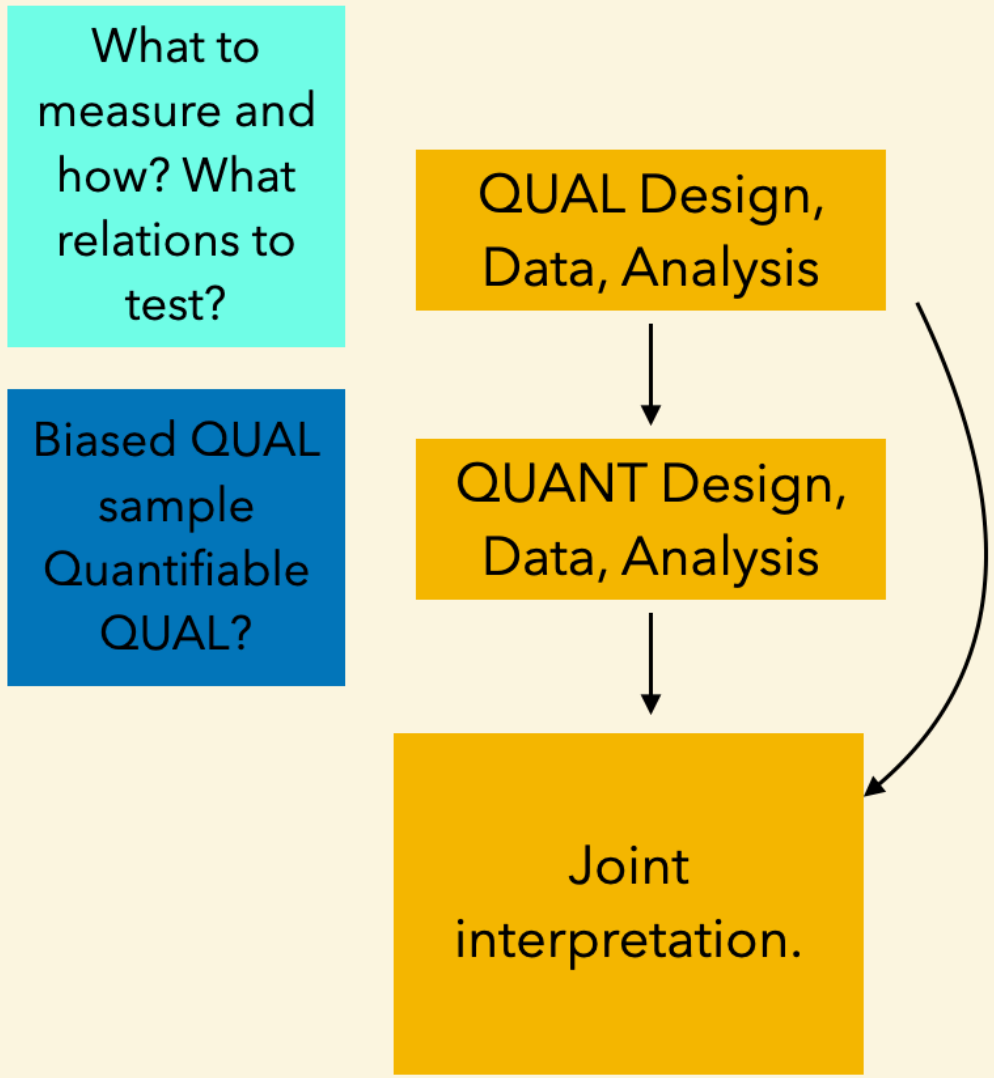


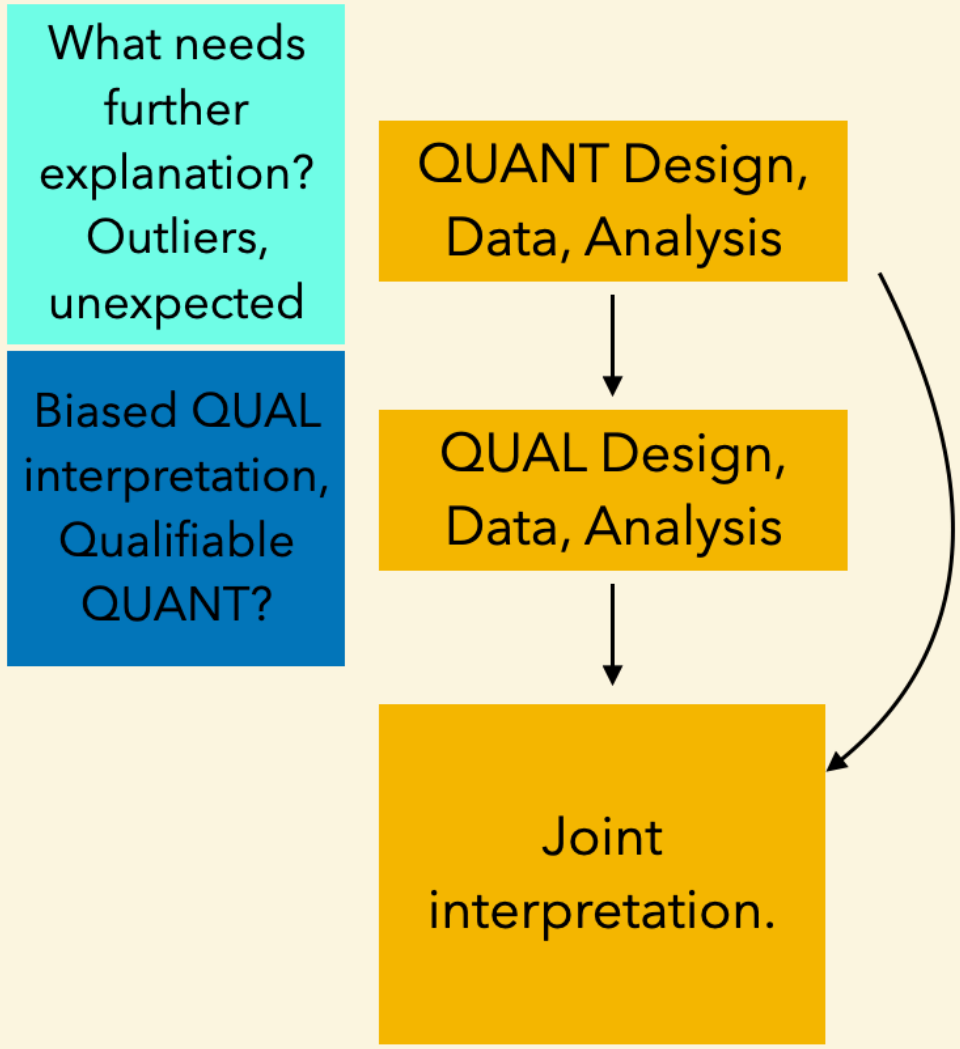
Con-current Design



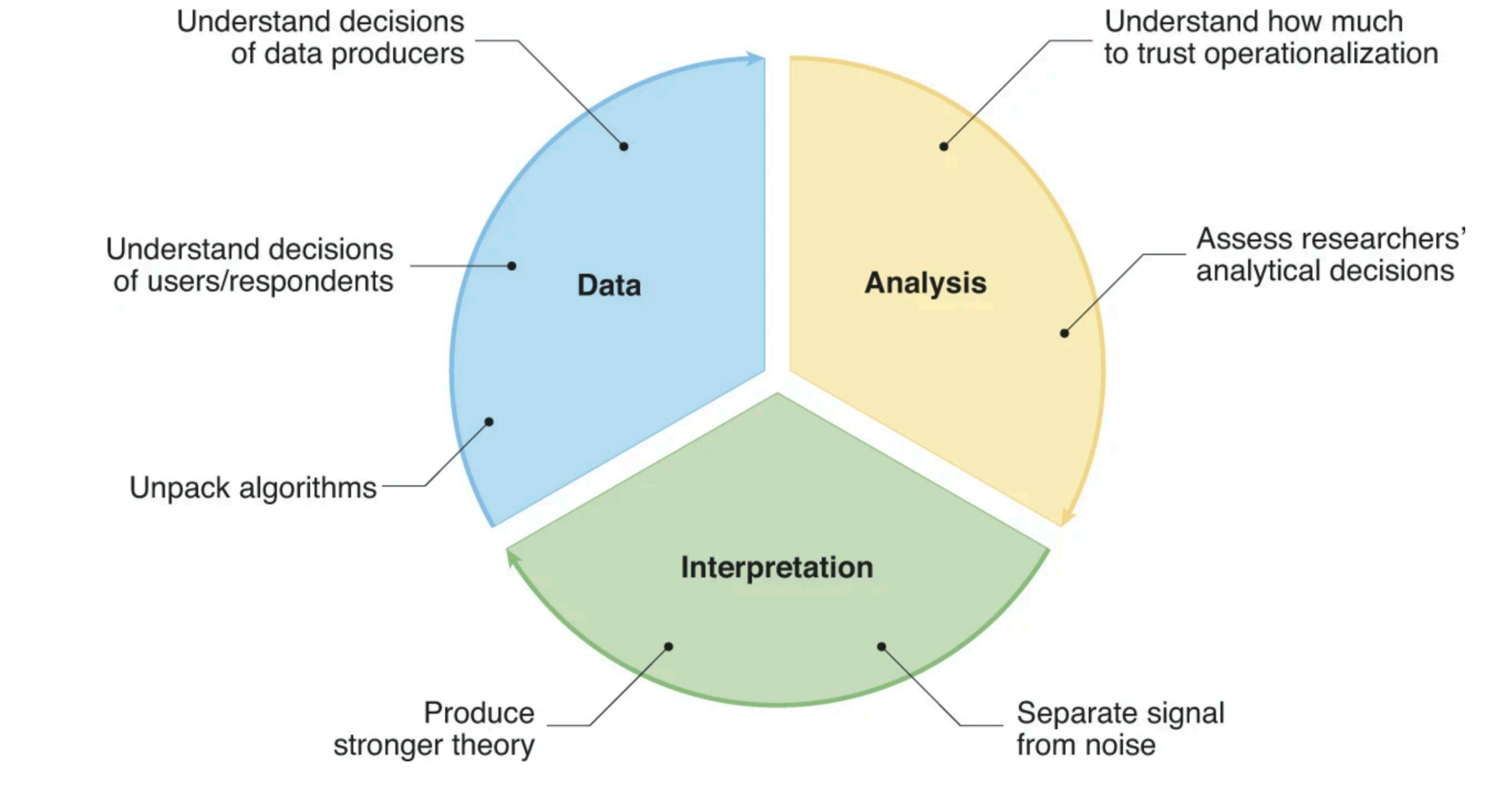
Sequencial Design




What to measure and how? What relations to test?
Biased QUAL sample Quantifiable QUAL?
QUAL Design, Data, Analysis
QUANT Design, Data, Analysis
Joint interpretation.


What needs further explanation? Outliers, unexpected
Biased QUAL interpretation, Qualifiable QUANT?
QUANT Design, Data, Analysis
QUAL Design, Data, Analysis
Joint interpretation.

What Role Does Qualitative Analysis Have in Social Data-Intensive Analysis?



The Role of Qualitative Research in the Data Generation Phase

- Decisions that affect data production
- Decisions made by respondents/users
- The nature of algorithms

The Role of Qualitative Research in the Analysis Phase

- Linking variables to concepts
- Analytical decisions

The Role of Qualitative Research in the Interpretation Phase

- From data to theory to data
- Recognizing noise

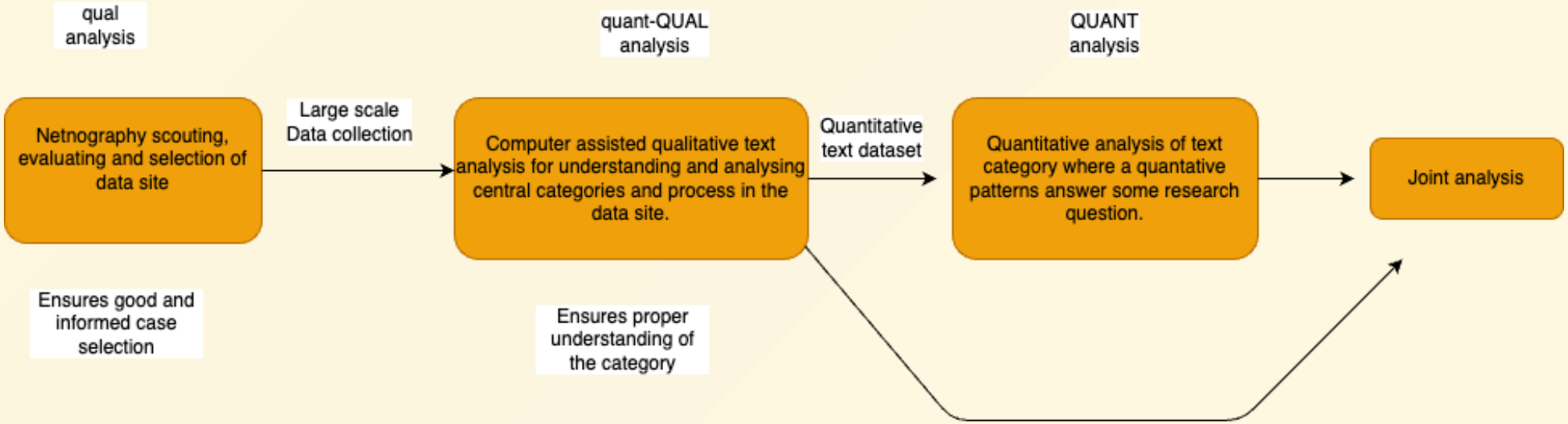
Offline vs Online Grounding with Qualitative Methods

- Sometimes, we need to move beyond online data to properly ground our analysis
- Many times, online observations provide a valuable resource for qualitative data
- New mixed methods approaches allow moving between qualitative and quantitative data
- Grigoropoulou & Small focus on the application of traditional qualitative methods

Cool Aspects of Digital Text Data and Context Analysis

- **Computational methods assist qualitative analysis**
  - Ensure saturation
  - Guard against bias in qualitative analysis
  - Enable discovery by alternative means
- **Qual and Quant Analysis of the Same Observations**
  - Ensure data integration
  - Ensure analytical integration (scale qualitative categories)
  - Allow us to revisit observations

Sequential Analysis Design (Typical in This Course)



Looking for Data Sites Over Easter?

- Start an immersion/research journal (this can be many different documents)
- Search for relevant data sites (more than one)
- Describe the relevant data sites in detail
- Focus on the data sites' relevance to your topic and their ability to support a mixed digital methods research design (qual and quant).

Exercise class

- Focus on: Topics and research questions
- Where to be: Look in the google sheet!

Next time: Find, evaluate and select datasite(s)